VEGI-192, a New Isoform of TNFSF15, Specifically Eliminates Tumor Vascular Endothelial Cells and Suppresses Tumor Growth

Wen Hou,[1] Dan Medynski,[2] Shirley Wu,[2] Xinli Lin,[2] and Lu-Yuan Li[1]

Abstract **Purpose:** We determined the antiangiogenic and anticancer activity of VEGI-192, a new isoform of TNFSF15 (VEGI, TL1), with a Lewis lung cancer murine tumor model.

Experimental Design: Recombinant human VEGI-192 was produced in *Escherichia coli* and purified to apparent homogeneity. The protein was given systemically via i.p., i.v., or s.c. injections to tumor-bearing C57BL/6 mice. Tumor growth rates, animal survival rates, and general toxicity were determined. Effect on endothelial cell/smooth muscle cell ratio of the tumor vasculature was analyzed.

Results: Systemic administration of VEGI-192 gave rise to a marked inhibition of tumor growth. As much as 50% inhibition of the tumor growth rate was achieved with treatment initiated when the tumor volumes reached nearly 5% of the body weight. Inhibition of tumor formation was also observed when VEGI-192 was given at the time of tumor inoculation. Consistently, we observed an increased survival time of the treated animals. The VEGI-192-treated animals showed no liver or kidney toxicity. The treatment eliminated tumor endothelial cells but not vascular smooth muscle cells, which remained associated with a residual vascular structure consisting of the basement membrane. In addition, we carried out immunohistochemical analysis of rat kidneys and found that vascular endothelial cell growth inhibitor (VEGI) expression is largely limited to endothelial cells.

Conclusions: Our findings indicate that VEGI is an endogenous inhibitor of angiogenesis, and that systemic administration of the VEGI-192 isoform resulted in inhibition of tumor angiogenesis and growth.

The recent success of applying an antiangiogenic agent Avastin (1) in clinical settings for cancer treatment provided the first set of evidence to support the hypothesis that inhibition of tumor neovascularization can bring significant benefit to cancer therapy (2). Because neovascularization under either physiologic or pathologic conditions is controlled by a balance of endogenous proangiogenic and antiangiogenic factors, an important approach to develop therapeutic agents for cancers and other angiogenesis-driven diseases is to use endogenous antiangiogenic factors (3).

We previously reported the discovery of an endothelial cell-specific gene product, vascular endothelial cell growth inhibitor (VEGI, TNFSF15), which exhibits 20% to 30% sequence homology to the tumor necrosis factor superfamily (4, 5). VEGI mRNA was found in many normal adult tissues, suggesting a physiologic role for this unique gene in the maintenance of the normal vasculature (6). We showed that VEGI is a potent and specific inhibitor of endothelial cell growth (4–6). VEGI exhibits two distinctly different activities on endothelial cells: growth arrest of G_0-G_1 cells and apoptosis of proliferating cells (7). These findings suggest that VEGI may have an important role in the regulation of vascular homeostasis.

There are three differential splicing variants of VEGI (7). The initially reported VEGI protein is composed of 174 amino acids (4, 5). Hydrophobic analysis predicted VEGI-174 to be a type II transmembrane protein, similar to most tumor necrosis factor (TNF) family members (8). Recombinant VEGI comprising only the putative extracellular domain exhibited effective inhibition of endothelial cell growth but had no effect on the proliferation of breast tumor cells or smooth muscle cells (4). Full-length VEGI-174 was found, however, to have no effect on tumor growth when overexpressed in cancer cells (5), whereas a secretable fusion protein (sVEGI) comprising a secretion signal peptide and the putative extracellular domain of VEGI-174 inhibited tumor growth when overexpressed in cancer cells (5). This indicates that a solubilized extracellular domain of VEGI is responsible for its biological activity. Two new isoforms, VEGI-251 and VEGI-192, were discovered subsequently (ref. 6; VEGI-251 is also known as TL1A; ref. 9).

We report here the anticancer activity of recombinant human VEGI-192 (Genbank accession no. AY434464) using a Lewis lung cancer (LLC) murine tumor model. To our knowledge, this study represents the first use of any form of purified VEGI protein in a preclinical cancer model. Previous attempts have been hindered by the difficulty of producing adequate quantities of VEGI by eukaryotic expression systems or by *Escherichia coli*. We found that systemic administration of the protein gave rise to a marked inhibition of tumor

Authors' Affiliations: [1]Department of Pathology, University of Pittsburgh Medical Center, Pittsburgh, Pennsylvania and [2]ProteomTech, Inc., Emeryville, California
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growth. The treatment led to specific elimination of endothelial cells but not of vascular smooth muscle cells. These findings, together with the observation that VEGI is largely associated with vascular endothelial cells in normal tissues, are consistent with the view that VEGI is an endogenous inhibitor of angiogenesis.

Materials and Methods

Cells. Adult bovine aortic endothelial cells were a gift from Dr. Peter Bohlen (Imclone Systems Inc., New York, NY) and cultured as described (6). Mouse LLC cells and human coronary artery smooth muscle cells were purchased from American Type Culture Collection (Manassas, VA) and cultured according to the vendor's instructions.

Preparation of recombinant VEGI-192. *E. coli* containing an expression plasmid was inoculated into 1.0 liter of Luria-Bertani broth containing ampicillin, induced with 500 μmol/L isopropyl-L-thio-B-D-galactopyranoside at $A_{600\ nm}$ = 0.6, and agitated for 3 hours at 37°C. The cells were collected by centrifugation, and the pellet subjected to freeze-and-thaw cycles. The inclusion bodies released were washed extensively with a buffer containing 50 mmol/L Tris, 100 mmol/L NaCl, 1% Triton X-100 (pH 8.0) and dissolved in a buffer containing 8 mol/L urea, 0.1 mol/L Tris, 1 mmol/L glycine, 1 mmol/L EDTA, 10 mmol/L β-mercaptoethanol, 10 mmol/L DTT, 1 mmol/L reduced glutathione, 0.1 mmol/L oxidized glutathione (pH 10) with a $A_{280\ nm}$ = 5.0. The solubilized inclusion bodies was refolded with a rapid dilution method as described (10–12). The refolded protein was concentrated by N_2-ultrafiltration and purified by size exclusion chromatography using Sephacryl S-300. The endotoxin concentration in the VEGI-192 preparation was 87 EU/mg.

Lewis lung carcinoma model. C57BL/6 black mice (Harlan, Indianapolis, IN) were injected s.c. on the flank with 1×10^6 LLC cells. The tumors were measured in a blinded manner with a dial caliper. The volumes were determined using the formula, volume = width × width × length × 0.52. The animals were randomized and divided into control and treatment groups before treatment. The treatment groups received recombinant human VEGI-192 via i.p., or s.c. (underneath the tumor) injections. The control groups received comparable injections of the vehicle. The animals were sacrificed at the end of each experiment. The tumors, other organs, and peripheral blood were collected for pathologic analysis. All experimental procedures were approved by The Institutional Animal Care and Use Committee at the University of Pittsburgh Medical Center.

Analysis of liver and kidney functions. Blood samples of vehicle- or VEGI-treated mice were analyzed by Antech (Lake Success, NY). Glucose (mg/d) was analyzed enzymatically using reagents from Synermed. Urea nitrogen (mg/d), creatine (mg/d), alanine aminotransferase (U/L), and phosphorous (mg/d) were assayed kinetically using dimethylacetamide reagents. Total protein (g/d) and total bilirubin (mg/d) were determined colorimetrically using reagents from Roche (Indianapolis, IN). All analytes were measured using a Hitachi-747 Spectrophotometric Chemistry Analyzer.

Immunohistochemistry. Tumors were fixed with 4% paraformaldehyde in PBS at 4°C for 4 hours, transferred to 30% sucrose (4°C) in PBS, and placed in ornithine carbamyl transferase compound on dry ice. Tumor sections (8-μm thickness) were subjected to immunostaining. Endothelial cells and vascular smooth muscle cells were identified, respectively, with a rat monoclonal antibody to CD31 (platelet/endothelial cell adhesion molecule 1; BD PharMingen (San Diego, CA), clone MEC 13.3) and a mouse monoclonal anti-α-SMA-FITC (Sigma, St. Louis, MO, clone 1A4). Vascular basement membrane was identified with a rabbit polyclonal antibody to type IV collagen (Cosmo Bio Co., Tokyo, Japan). Cell nuclei were stained with Hoechst (Sigma). Secondary antibodies were biotinylated anti-rat IgG (Vector Laboratories, Burlingame, CA), rabbit anti-rat IgG-TRITC (Sigma), donkey anti-rat IgG-FITC (The Jackson Laboratory, Bar Harbor, Maine), and goat anti-rabbit IgG-TRITC (Sigma). AMCA avidin D (Vector Laboratories) was used for biotin detection. Avidin-biotin complex standard kits and 3,3′-diaminobenzidine kits (Vector Laboratories) were used according to manufacturer's instructions.

Kidneys from female rats were fixed with 10% formalin and embedded in paraffin. Serial sections (5 μm) were subjected to immunostaining with either mouse anti-human VEGI monoclonal antibody (3-12D) we developed, mouse anti-rat CD31 (platelet/endothelial cell adhesion molecule 1) monoclonal antibody TLD-3A12 (BD PharMingen), or mouse anti-rat vascular endothelial growth factor (VEGF, C-1) sc-7269 (Santa Cruz Biotechnology, Santa Cruz, CA) followed by a biotinylated anti-mouse antibody and streptavidin-conjugated peroxidase (VECTASTAIN Elite avidin-biotin complex reagent, Vector Laboratories). The specificity of antibody 3-12D was verified by using purified VEGI-192 to block 3-12D binding to VEGI in both Western blotting and immunostaining.

Microscopy and image analysis. The specimens were examined with a Nikon Eclipse E800 fluorescence microscope equipped with single, dual, and triple fluorescence filters and a low-light, RETIGA 1300C CCD Camera (Quantitative Imaging Corp., Burnaby, British Columbia, Canada) with Qcapture software. Images were saved as digital files. Image analysis was carried out with Image-pro plus software (Media Cybernetics, Inc., San Diego, CA) or Image-J (NIH).

Results

Vascular endothelial cell-specific localization of vascular endothelial cell growth inhibitor. We have developed a mouse monoclonal antibody (3-12D) against a recombinant human VEGI protein consisting of the peptide segment common to all three isoforms and have used the antibody to determine tissue distribution of VEGI by immunohistochemistry in rat kidney (Fig. 1), because kidney is the organ with the most abundant VEGI mRNA (5). The results indicate that VEGI is specifically associated with vascular endothelial cells in the veins and arteries, in a pattern that is nearly identical to that of CD31 (platelet/endothelial cell adhesion molecule 1), an endothelial cell marker, or that of VEGF-A, a growth factor whose receptors are specific to endothelial cells. Because this antibody was raised against a peptide common to all VEGI isoforms, the distribution pattern shown here represents that of all VEGI isoforms. These results show that VEGI is an endogenous factor produced by endothelial cells under physiologic conditions.

Inhibition of tumor formation. We determined the anticancer activity of systemically given recombinant VEGI-192 with a LLC murine tumor model using immunologically intact C57BL/6 black mice. Recombinant VEGI-192 causes a dose-dependent inhibition of the growth of adult bovine aortic endothelial cells *in vitro*, whereas leaving that of human coronary artery smooth muscle cells or LLC cells unaffected (Fig. 2A), suggesting that inhibition of LLC tumorigenesis by VEGI should result from inhibition of angiogenesis. S.c. implanted LLC cells formed rapidly growing tumors. We first determined the effect of the systemically given recombinant VEGI-192 (5 mg/kg) on tumor formation rate. We found that i.p. administration of VEGI-192 at the time of the cancer cell inoculation resulted in a marked inhibition of tumor formation (Fig. 2B). When the tumor volumes were assessed on day 5 after inoculation, all the animals in the vehicle-treated group had developed s.c. tumors (five of five or 100%), with a mean volume (±SD) of 35 mm^3, whereas only one third of the VEGI-treated group exhibited measurable tumors (two of six or 33%). The results indicate

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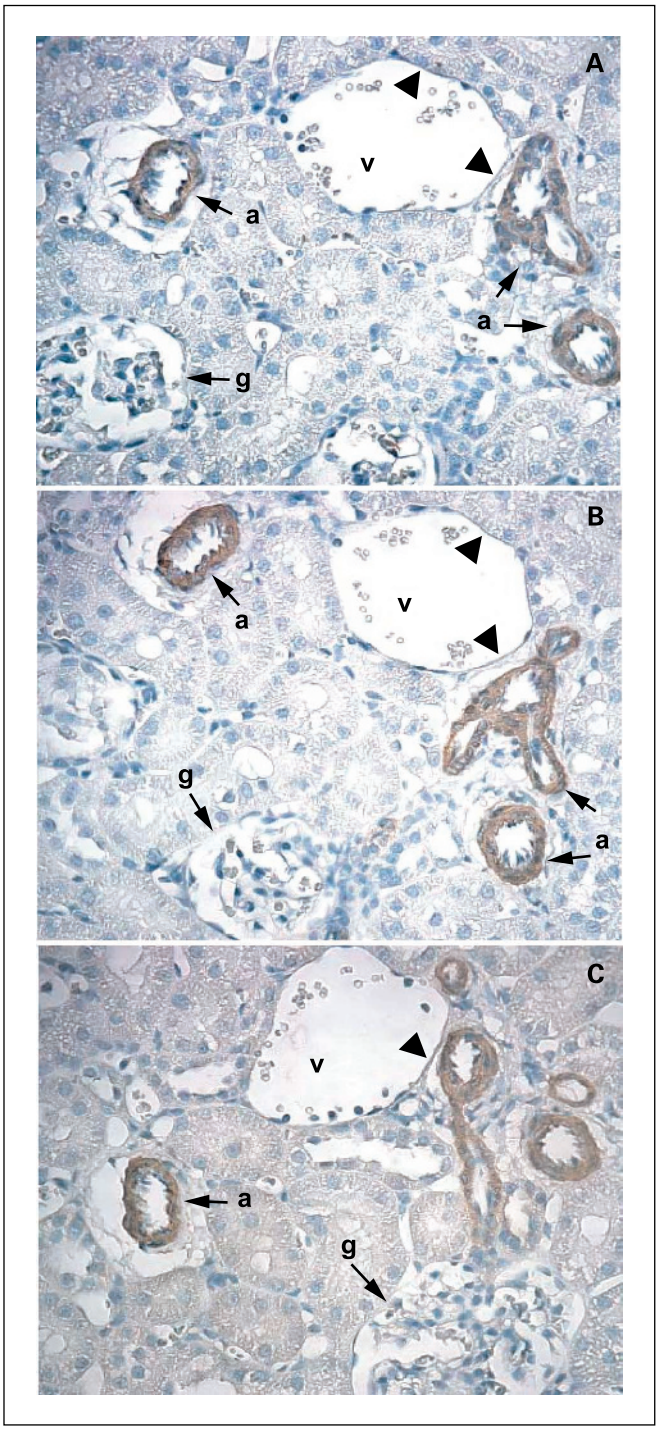

Fig. 1. Vascular localization of VEGI in kidney. Sections (5 μm) of formalin-fixed, paraffin-embedded rat kidney were subjected to immunostaining with a mouse monoclonal antibody (3-12D) against human VEGI (*A*) and compared with the immunostaining patterns of CD31 (*B*) and VEGF (*C*). a, artery; g, glomerulus; v, vein. *Arrowheads,* positive endothelial cell staining (*dark brown*) in the vein.

that systemic administration of VEGI led to the retardation of tumor formation by the cancer cells.

Inhibition of the growth of established tumors. We then determined the ability of VEGI-192 to inhibit the growth of established tumors. In one experiment, the treatment was initiated as soon as the tumors were palpable (Fig. 2C). The animals were treated on days 5, 9, and 12 by i.p. administration of VEGI-192 (5 mg/kg). The control group was treated with vehicle. A significantly slower tumor growth rate was observed for the VEGI-treated group. In another experiment, the treatment was initiated when the tumors reached about 5% of the body weight (Fig. 2D). The animals were treated twice at a dosage of 5 mg/kg on days 11 and 14. We observed about 60% decrease of the tumor growth rate within 1 week. Comparable inhibition of the tumor growth rates was obtained using either i.p. or s.c. (intratumoral, i.t.) treatments. These data strongly suggest that systemically delivered VEGI was able to inhibit the growth of established tumors.

Enhancement of the survival of tumor-bearing animals. We determined the effect of VEGI treatment on the survival of the tumor-bearing animals (Fig. 3). Recombinant VEGI-192 was given by i.v., i.t., or i.p. injections (5 mg/kg) on days 11 and 14. The control group was treated with the vehicle. The animals were sacrificed once the tumor volume reached 4,000 mm^3. We found that regardless of the administration route, VEGI treatment gave rise to significantly increased survival time of the tumor-bearing animals, as the median survival time for the untreated group was 17 days, whereas that for the VEGI-treated group was 22 days, representing a nearly 30% increase of the median survival time.

Absence of liver and kidney toxicity resulting from vascular endothelial cell growth inhibitor treatment. We determined the effect of systemically given VEGI-192 on liver and kidney functions. The animals were treated by i.p. injection of recombinant VEGI-192 at 20 mg/kg daily for 5 days. The blood was collected 24 hours after the last injection. The liver and kidney functions were determined by measuring serum levels of a number of relevant enzymes and other factors (Fig. 4). No significant difference was observed between the control groups and the VEGI-192 treated groups. These results indicate that VEGI-192 treatment did not cause significant adverse effect on the function of the liver and kidney for the duration of the experiment.

Specific eradication of endothelial cells by VEGI-192. We determined the effect of VEGI-192 treatment of the tumor-bearing mice on the abundance and structure of the tumor blood vessels. Freshly frozen tumors were sectioned and subjected to immunostaining for endothelial marker CD31 (red) and α-smooth muscle cell antigen (SMA, green; Fig. 5A-B). We analyzed 15 fields on each slide that contained the greatest number of microvessels ("hotspots") by computer-assisted image analysis. The densities of the red or green pixels per field (400× magnification) were determined (Fig. 5C). The density of the endothelial cells, measured as the total pixels occupied by CD31-positive cells, exhibited an 88% decrease within 1 week of treatment and a further decrease within 3 weeks. Interestingly, the number of the smooth muscle cells remained relatively unchanged. Thus, the ratio of endothelial cells to smooth muscle cells decreased markedly in VEGI-treated tumors, changing from 1.8 to 0.4 and 1.8 to 0.15 after the animals had been treated twice a week for 1 or 3 weeks, respectively. We carried out a similar analysis for another endothelial cell marker, CD105, and obtained identical results (Fig. 5D). We also immunostained the tumor sections for desmin and vimentin, markers of mesenchymal cells that are precursors of vascular smooth muscle cells, and found that VEGI-192 treatment had no effect on the number of these cells

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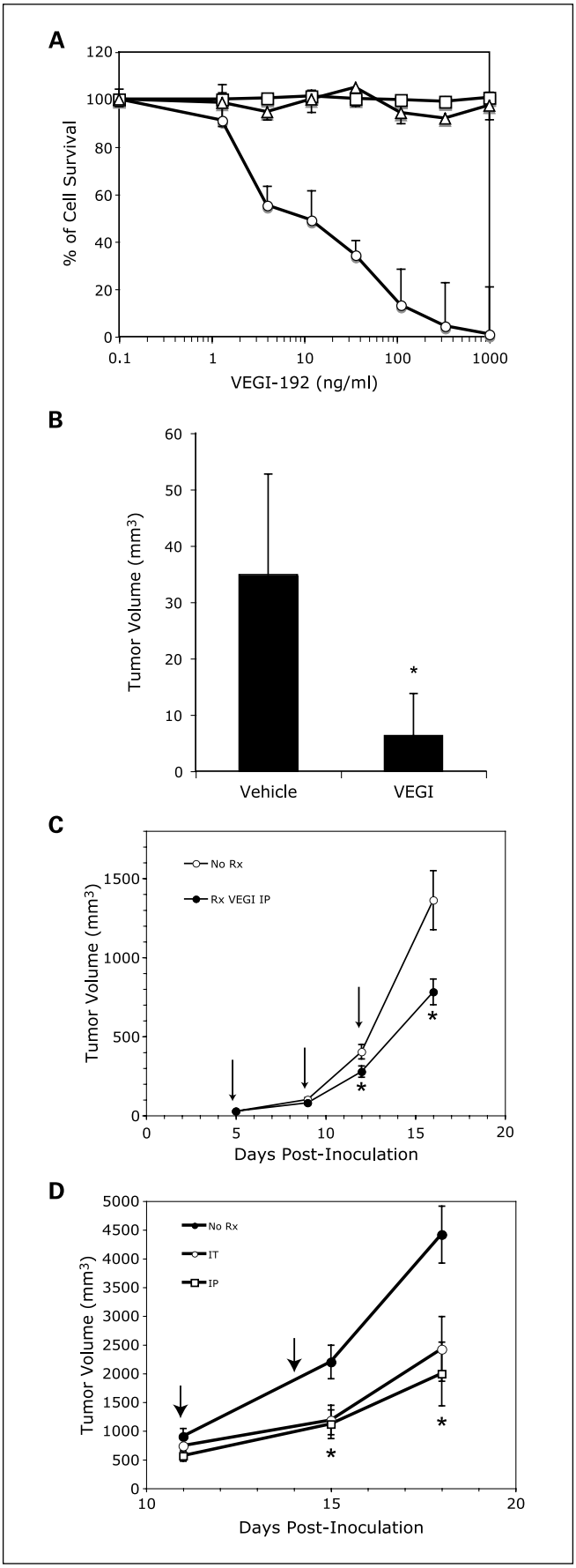

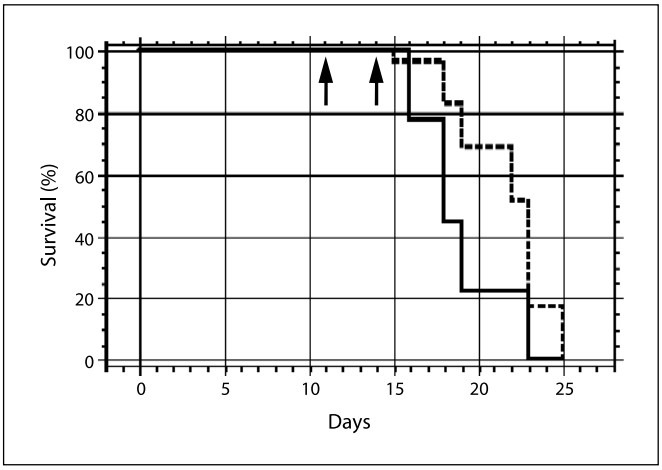

Fig. 3. VEGI treatment of tumor-bearing mice (LLC tumors on C57BL mice) enhanced the survival of the animals. VEGI (5 mg/kg) was given on days 11 and 14. Treated group (*dashed line*), $n = 29$ (pooled from i.p. = 10, i.v. = 10, i.t. = 9). Untreated group (*solid line*), $n = 9$. $P = 0.0128$ (log-rank test).

in the tumors (Fig. 5E). These results indicate that the VEGI treatment caused specific eradication of tumor vascular endothelial cells.

Residual vascular structure without endothelial cells. We determined the effect of VEGI-induced elimination of vascular endothelial cells on the residual vascular structure. By immunostaining for markers of endothelial cells (CD31), vascular smooth muscle cell, and blood vessel basement membrane (collagen IV), we found that the residual vascular structures consisted of basement membrane of previously existing blood vessels (Fig. 6). The residual vascular structures in the VEGI-treated tumors were largely depleted of endothelial cells (Fig. 6A). Some contained smooth muscle cells (Fig. 6B). The blood vessels in the untreated tumors of the control groups contained endothelial cells, smooth muscle cells, and the basement membrane (Fig. 6C-D). Longitudinal sections of tumor blood vessels are also shown here to further show the elimination of endothelial cells, but not of smooth muscle cells, from the tumor vasculature by VEGI treatment (Fig. 6E-H). These data indicate that elimination of endothelial cells in tumor blood vessels does not necessarily lead to the immediate and complete eradication of the tumor vasculature.

Fig. 2. Inhibition of LLC tumor formation and growth. *A,* recombinant VEGI-192 inhibits the growth of adult bovine aortic endothelial cells but not that of LLC or coronary artery smooth muscle cells in culture. Adult bovine aortic endothelial (○), LLC (□), coronary artery smooth muscle (△). *B,* inhibition of LLC tumor formation. LLC cells (1×10^6 per injection per animal) were inoculated on the flank of a C57BL black mouse on day 0. The animals were treated by i.p. injection of recombinant VEGI-192 (5 mg/kg) immediately following cancer cell inoculation. The treatment was repeated daily until day 4 when the tumor volumes were determined. *, $P < 0.002$ (*t* test; untreated, $n = 5$; treated, $n = 6$). *C,* inhibition of the growth of newly implanted LLC tumors. LLC cells (1×10^6 per injection) were inoculated on the flank of a C57BL/6 black mouse on day 0. Recombinant VEGI (20 mg/kg) was given on days 5, 9, and 12 (*arrows*) by i.p. injection. Tumor volumes were measured immediately before VEGI treatment. *, $P < 0.05$ (*t* test; treated groups $n = 9$, untreated group $n = 9$). *D,* inhibition of the growth of established LLC tumors. LLC cells (1×10^6 per injection) were inoculated on the flank of a C57BL/6 black mouse on day 0. Recombinant VEGI (5 mg/kg) was given on days 11 and 14 (*arrows*) by either i.t. (*IT*) or i.p. (*IP*) injection. Tumor volumes were measured on days 15 and 18. *, $P < 0.05$ (*t* test; treated groups $n = 9$, untreated group $n = 9$).

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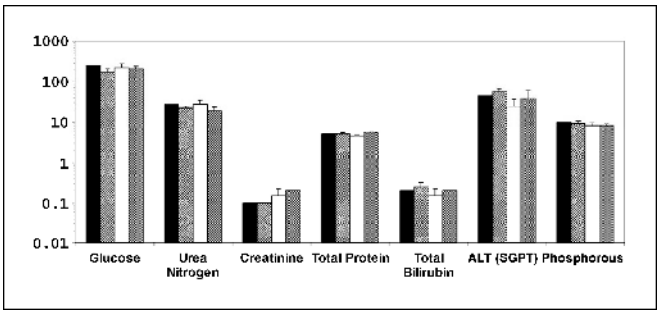

Fig. 4. Standard chemistry analyses in the blood of tumor-free PBS-treated mice (*black columns, n* = 2); tumor-free VEGI-treated mice (*striped columns, n* = 2); tumor-induced PBS-treated mice (*white columns, n* = 2); and tumor-induced VEGI-treated mice (*hatched columns, n* = 3). The units (*y*-axis) are, respectively, glucose, mg/d; urea nitrogen, mg/d; creatine, mg/d; total protein, g/d; total bilirubin, mg/d; alanine aminotransferase (*ALT*), U/L; phosphorous, mg/d.

Discussion

We showed, using the murine LLC model, that systemically delivered recombinant VEGI-192 exhibited potent inhibitory activity on tumor formation as well as tumor growth. In one experiment, we treated the animals by i.p. injection with recombinant VEGI at the time when the cancer cells were implanted. Marked inhibition of tumor formation was observed with the treated group. In another experiment, we allowed the tumor to reach about 5% of the body weight before the animals were treated. A substantially retarded growth of the tumors was observed for the treated group during the ~1-week period following the treatment compared with the tumor growth rate of the untreated group. This result is highly significant because similar inhibition of tumor growth was obtained when VEGI was injected directly into the base of the tumors, suggesting that the effect of VEGI was systemic. Furthermore, there was a significant improvement of the survival time of the tumor-bearing animals resulting from VEGI treatment. Importantly, the VEGI-treated animals showed no signs of liver, kidney, or overt toxicity. These findings suggest that recombinant VEGI is a potential therapeutic agent for the treatment of cancer.

We noticed that the most effective anticancer activity of VEGI was seen within the first week of treatment. It is plausible that the recombinant protein was not sufficiently stable to give a more sustained activity or, alternatively, that an antigenic reaction may have led to the quick clearance of the human

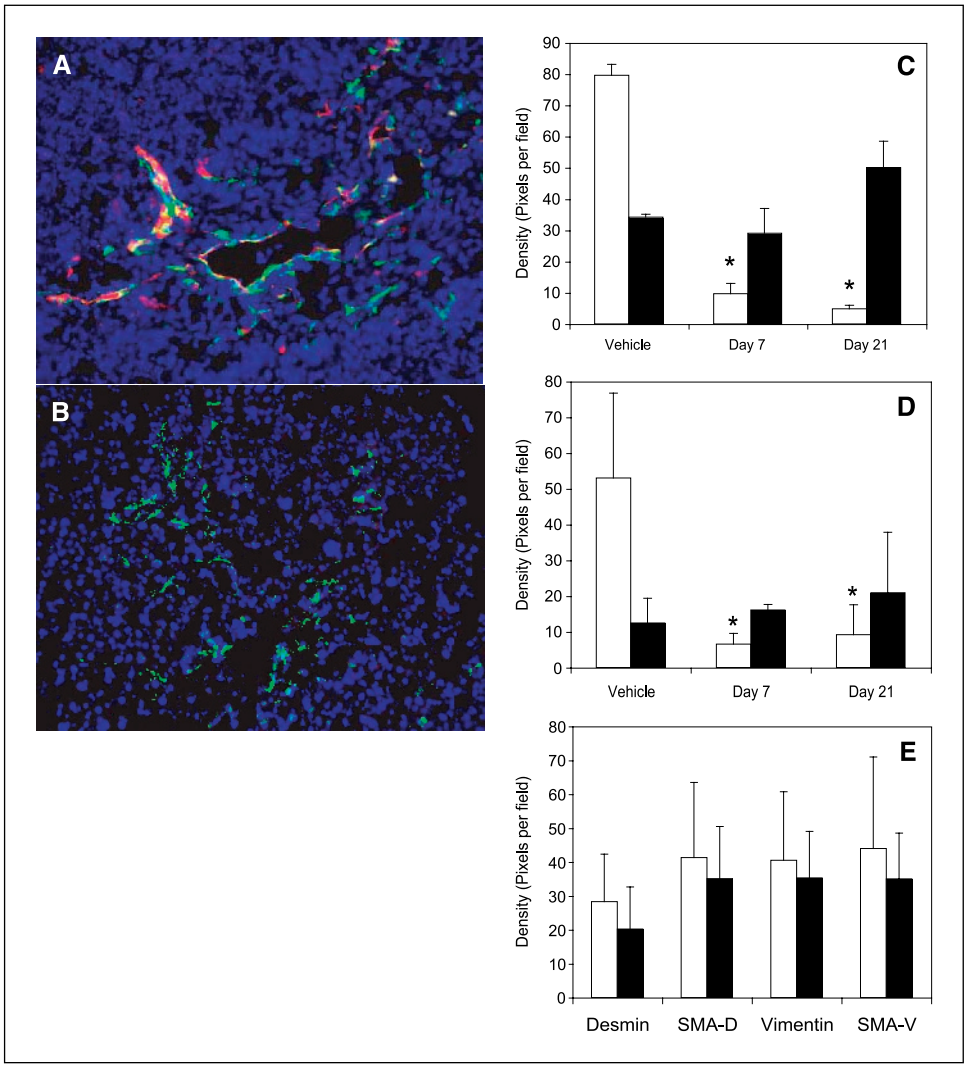

Fig. 5. Specific elimination of endothelial cells by VEGI in LLC tumors. Tumors were retrieved at the end of the experiment (3 weeks) from VEGI-treated animals and vehicle-treated controls and processed as described in Materials and Methods. The tumor volumes were 2,000 to 4,000 mm^3. Sections taken from tumor areas where there was no apparent necrosis were subjected to fluorescent immunostaining. Endothelial cells and smooth muscle cells were identified with specific markers CD31 (*red*) and SMA (*green*), respectively. *Blue staining,* nuclei. *A,* image of a typical tumor section from vehicle-treated group. Magnification, 200×. *B,* image of a typical tumor section from VEGI treated group. Magnification, 200×. *C,* quantitative analysis of CD31-positive (*red*) and SMA-positive (*green*) areas of the tumor sections. *White columns,* CD31-positive endothelial cells. *Black columns,* SMA-positive smooth muscle cells. *D,* quantitative analysis of CD105-positive (*red*) and SMA-positive (*green*) areas of the tumor sections. *White columns,* CD105-positive endothelial cells. *Black columns,* SMA-positive smooth muscle cells. *E,* comparison of densities of smooth muscle cells stained for SMA with those stained for desmin or vimentin, markers of precursors of vascular smooth muscle cells. SMA-D, costaining of SMA with desmin. SMA-V, costaining of SMA with vimentin. *White columns,* untreated tumors. *Black columns,* VEGI-192-treated tumors. *, $P < 0.01$ (*t* test) between vehicle and VEGI treated groups for CD31 staining (five animals per group; 15 areas per section analyzed).

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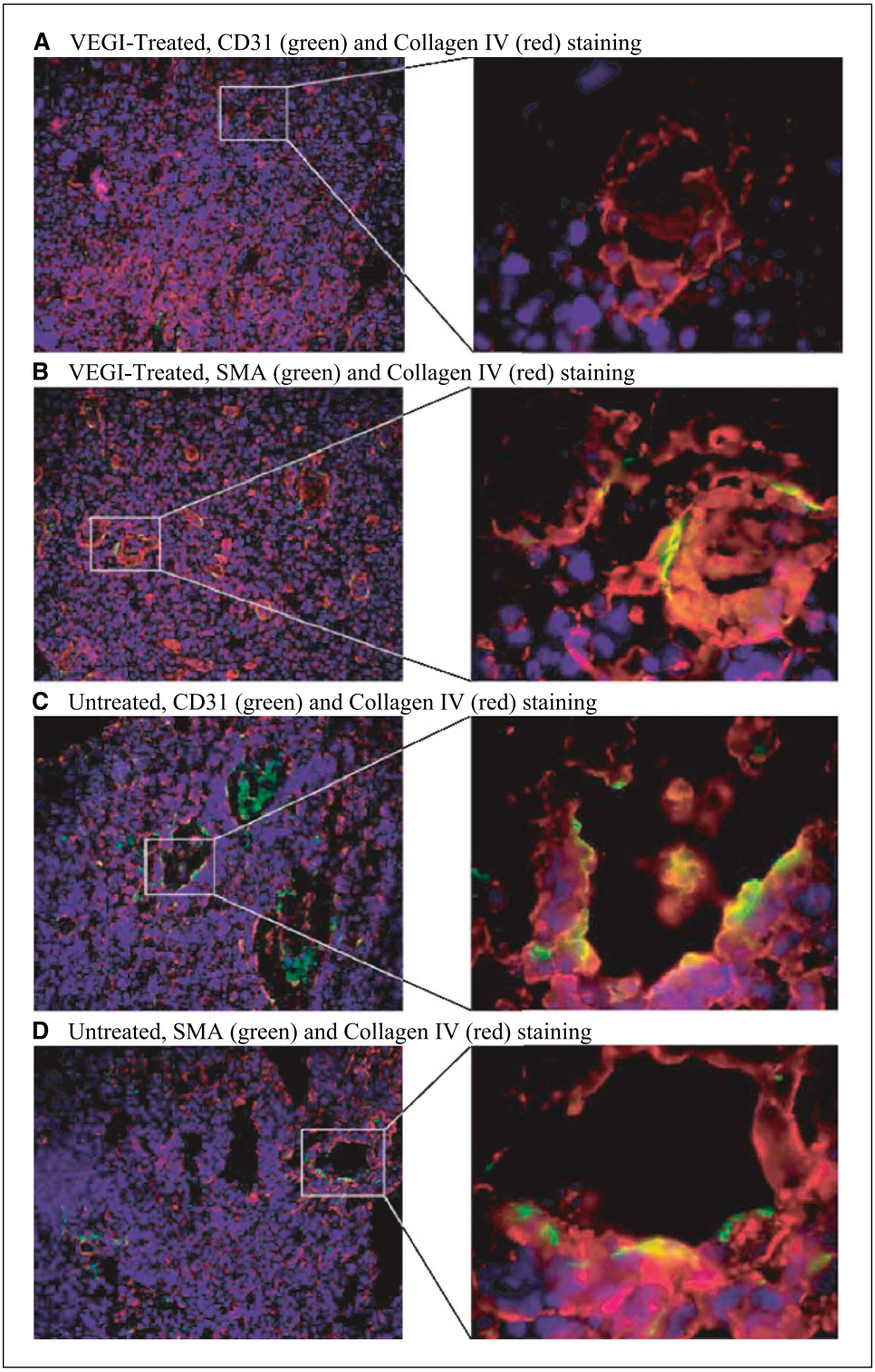

Fig. 6. Images of VEGI-treated and untreated control LLC tumors immunostained for endothelial cells (CD31), smooth muscle cells (SMA), and blood vessel basement membrane (collagen IV), indicating the existence of ghost vessels in VEGI-treated LLC tumors. *Blue staining,* cell nuclei. *A,* VEGI-treated, CD31 (*green*), and collagen IV (*red*) double staining; notice the lack of CD31-positive cells. *B,* VEGI-treated, SMA (*green*), and collagen IV (*red*). *C,* untreated, CD31 (*green*), and collagen IV (*red*). *D,* untreated, SMA (green), and collagen IV (*red*).

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protein in the mice. This will be investigated as we move to determine the pharmacologic properties of recombinant VEGI-192.

Our results indicate that the antiangiogenic activity of VEGI-192 specifically targets endothelial cells not the other cellular components of the vasculature such as smooth muscle cells. We reported previously that VEGI is a specific inhibitor of endothelial cell growth *in vitro* and a potent inhibitor of blood vessel growth with angiogenesis models, including the chicken chorioallantoic membrane angiogenesis model (4) and the Matrigel implant model (5). We show here that VEGI treatment of the tumor-bearing mice results in a specific elimination of endothelial cells in tumors. The density of endothelial cells in the tumor vasculature exhibited a nearly 90% decrease

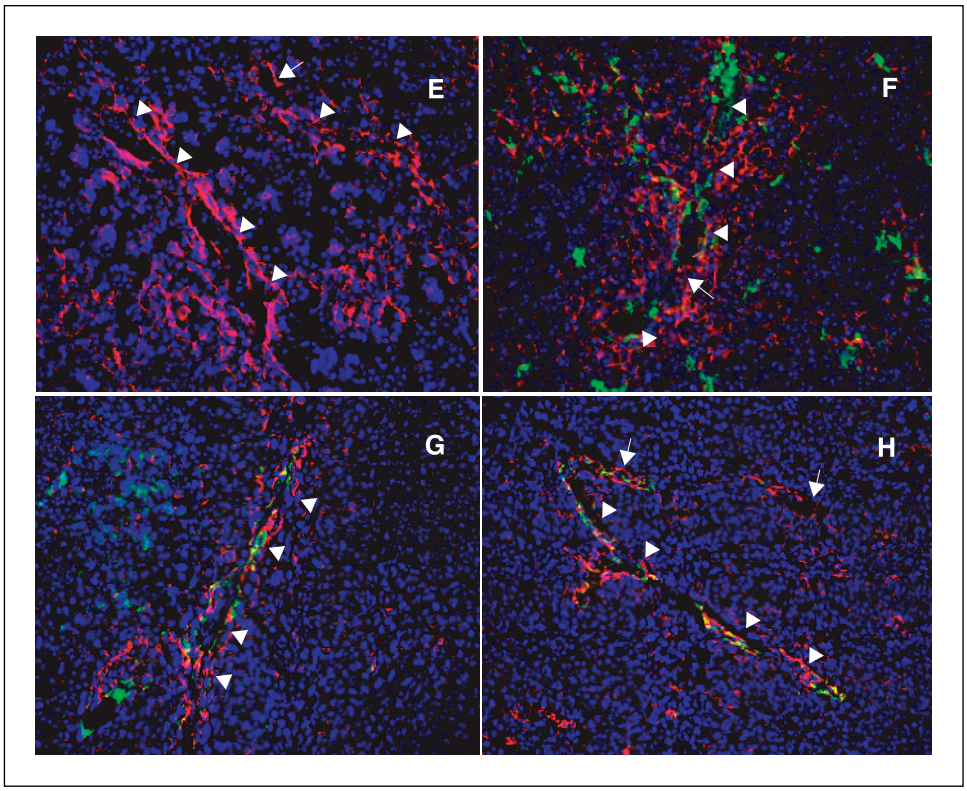

Fig. 6 *continued*. *E-H*, longitudinal blood vessels (*arrowheads*). CD31 (*green*) and collagen IV (*red*) double-stained sections of VEGI-treated (*E*) or untreated tumors (*F*). SMA (*green*) and collagen IV double-stained sections of VEGI-treated (*G*) or untreated (*H*) tumors.

compared with that in the tumors of the untreated group. VEGI treatment had no effect on vascular smooth muscle cells or their precursors, as the density of these cells remained basically unchanged. Therefore, the eradication of endothelial cells by the systemically given VEGI was highly specific.

Our data also showed a persistent existence of the residual vascular structures following the elimination of vascular endothelial cells by VEGI treatment. It is not yet clear whether the residual vascular structures would support blood circulation in the tumors. It is plausible, however, that the residual vascular structure may provide a framework or foundation for the recruitment of circulating endothelial progenitor cells to rebuild the blood vessels in the tumor, as it is now well documented that postnatal vasculogenesis is an important process in tumor neovascularization (13, 14). Our findings confirmed what was described by McDonald et al. (15). Those authors reported the presence of "ghost vessels" in LLC treated with antiangiogenic agents AG013736 or VEGF-Trap that inhibited VEGF signaling. They found that blood vessel basement membrane in the treated tumors, as marked by collagen IV, persisted after endothelial cells degenerated.

Migone et al. (9) reported that a truncated preparation of one of the VEGI isoforms, VEGI-251, which they named TL1A and in which a putative secretion signal peptide was removed, was able to bind to death receptor-3 (TNFRSF25), and induce nuclear factor-κB activation and apoptosis in death receptor-3–expressing cell lines; however, this preparation was unable to inhibit endothelial cell growth. We are currently investigating whether death receptor-3 is responsible for the activity of VEGI-192. In addition, a number of investigators have reported that TL1A plays an important role in inflammation and hematopoiesis (16–19). We are also investigating similarities and differences in the activities of VEGI-192 and TL1A in this regard.

We showed previously that VEGI mRNA is readily detectable in a variety of human organs and tissues (5). We show here with kidney as an example that the expression pattern of VEGI is highly similar to that of endothelial cell marker CD31 as well as VEGF, of which the receptors are specific to endothelial cells. Our data also indicate that systemically given VEGI does not adversely affect the functions of liver and kidney under the experimental conditions, suggesting that VEGI treatment of the animals did not damage the endothelial cells in the normal, quiescent vasculature, as it did to the proliferating endothelial cells in the tumor vasculature. We reported previously that G_0-synchronized endothelial cells were unable to reenter the growth cycle in the presence of VEGI (7). These findings support our view that VEGI plays a critical role in the maintenance of the quiescence of a normal vasculature.

In summary, our data strongly suggest that VEGI is an endogenous inhibitor of neovascularization, and that recombinant VEGI-192 is a potentially valuable anticancer agent as it is capable of eliminating angiogenic endothelial cells in tumors when systemically administrated to LLC tumor-bearing animals. This agent inhibits both the initiation of tumors and the growth of established tumors. Furthermore, treatment with VEGI is nontoxic to the host, at least in short-term treatment settings.

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